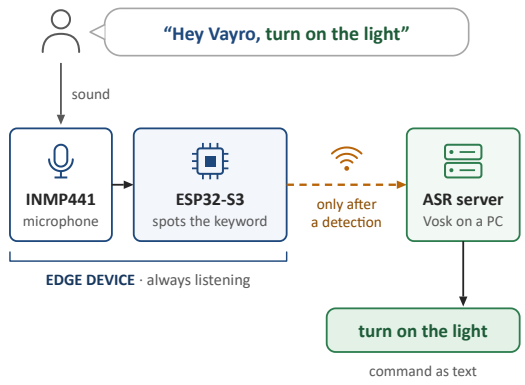
Hey Vayro

Voice control that listens on a small chip, and sends audio only when you call it.

Commercialization snapshot
Team DATA_Edgepulse
SIH 2026 · PS 26172 (ISRO)

1 PROVEN TODAY working prototype, measured on real hardware



No audio leaves the room until it wakes up

245 KiB
peak memory · PS limit 256 KB

9.3%
CPU while listening · PS limit 10%

138 ms
from wake word to the server (median)

20 / 25
wake-ups by a voice it was never trained on

2 WHAT WE WILL FIX NEXT engineering work, not dead ends

False wake-ups
TODAY about 16 an hour with TV and talk
TARGET about 1 an hour

More voices, more homes
TODAY 2 speakers, 1 home tested
TARGET 10+ speakers, several homes

Ready to install
TODAY fixed settings, text output only
TARGET phone set-up, encryption, controls real devices

3 WHO NEEDS IT

Indian device makers
Their own "Hey <Brand>" wake word, without a vendor licence or a huge data collection.

Labs and campuses
Hands-free voice for gloved hands, with audio kept on the campus server.

Privacy-minded homes
Voice control of the home, with no vendor cloud listening in.

4 HOW IT BECOMES A BUSINESS

DONE
Prototype
Full chain works within the PS limits

NEXT
Robust product
Fewer false wake-ups, more voices, secure set-up

PLANNED
Pilots
A campus lab, a few homes, one device maker

FUTURE
Scale
Wake-word service, kits and support plans

FIRST REVENUE
Custom wake words
One-time fee per brand word, with a measured test report

Kits and installs
Edge devices (about Rs 2,000 in parts each) plus a local server

Support plans
Yearly model and firmware updates, maintenance

Integration
Home and building automation, Indian-language speech

Open core: the firmware and model stay open source; customers pay for what they cannot easily do themselves. We will not run a cloud that processes customers' audio, and will never sell voice data.

Beyond SIH
SIH is our first validation, not the finish line. Whatever the result, we will keep cutting false wake-ups, run real pilots and build a product people can install, in the open.
github.com/Kartikinfinity/vayro-edge-voice-activator

Metrics measured on our final firmware, Sep 2026; full evidence in the technical report on GitHub. Targets and future stages are plans, not results. No customers or revenue yet.